AIR-GAPPED FORENSIC ACCOUNTING WORKSTATION • ZERO TELEMETRY

VaultSheet Air-Gap: Quick-Start Guide

Step-by-step operating manual for standalone PDF-to-Excel statement conversion and forensic B2B/GST auditing.

What is VaultSheet Air-Gap? VaultSheet is a 100% offline, privacy-first desktop financial workstation built for Chartered Accountants (CAs) and corporate finance teams. It operates in two flexible modes: **(1) Standalone High-Speed PDF-to-Excel Converter** for any bank statement, and **(2) Full Forensic & B2B GST Contract Auditor** with TallyPrime XML voucher generation.

1. End-to-End Architecture & Data Flow



Figure 1: All decryption, parsing, and auditing occur strictly in volatile memory (RAM) on localhost (127.0.0.1).

2. Step-by-Step Operating Workflow (5 Simple Steps)

1. **Launch the Workstation:** Double-click `start_vaultsheet.bat` (or run `streamlit run app.py`). The app opens locally at `http://127.0.0.1:8501` with outbound internet sockets blocked for total data privacy.
2. **Load a Bank Statement PDF (Sidebar Step 1):** Upload any digital, borderless, or scanned bank statement PDF (or pick a synthetic demo file). If the PDF is encrypted, type the password for instant RAM-only decryption.
3. **B2B Vendor Master CSV (Sidebar Step 2 — Optional):** Leave this empty for rapid PDF-to-Excel conversion. Or upload a custom vendor contract CSV—VaultSheet's *Universal Schema Auto-Mapper* automatically maps your column names.
4. **Execute Pipeline (Sidebar Step 3):** Optionally toggle *Tier-2 Local Ollama AI* for smart narration tagging, enter your Tally ledger name, and click **Run Forensic Audit Pipeline**.
5. **Review, Edit & Export:** Inspect the 5 workspace tabs, make live cell edits if needed, download your **Multi-Sheet Excel Workbook** or **TallyPrime XML**, and click **Purge RAM & Reset Session** when done.

3. Understanding the 5 Interactive Workspace Tabs

Once the pipeline finishes, VaultSheet organizes your parsed statement and audit findings into five specialized tabs:

WORKSPACE TAB	WHAT IT SHOWS	KEY ACTION / BENEFIT
01 · Master Audit Ledger	Interactive 15-column ledger with Dates, Narrations, Debits, Credits, Balances, and Category tags.	Edit any cell directly on screen; running balances recalculate and re-verify automatically.
02 · Forensic Risk & Fee Audit	Flags math errors, same-day duplicate debits, hidden bank fees, 2.5σ outliers, and round-sum outflows.	Isolates suspicious transactions and bank overcharges immediately without manual ticking.
03 · B2B Contract & GST Recon	Matches statement debits against your Vendor Contract Master CSV and checks 18% GST + TDS math.	Automatically skipped during standalone PDF-to-Excel mode; flags vendor overbilling when enabled.
04 · Cash Flow & Spend Analytics	Visualizes inflows vs. outflows, category-wise spend concentration, and daily balance trends.	Gives CFOs and auditors an instant visual summary of liquidity and burn rate.
05 · Enterprise Exports	One-click download center for formatted Excel (`.xlsx`), TallyPrime (`.xml`), and JSON.	Eliminates manual voucher entry in TallyPrime and produces client-ready Excel audit workbooks.

4. Under the Hood: Extraction Modes & The 6 Forensic Rules

3-Mode Self-Healing PDF Parser (`parser_engine.py`):

- **Mode A (`native_grid`):** Extracts vector tables from standard digital statements and stitches multi-line narrations.
- **Mode B (`native_stream`):** Uses horizontal Y-coordinate clustering for modern borderless bank PDFs.
- **Mode C (`offline_ocr`):** Converts scanned image PDFs at 200 DPI using `pypdfium2` + `RapidOCR` completely offline.

6-Rule Forensic Engine (`categorizer_and_auditor.py`):

- **Rule 1 (Balance Healing):** Verifies `Prev Bal - Dr + Cr = Curr Bal` and heals swapped debit/credit columns.
- **Rules 2 & 3 (Duplicates & Fees):** Traps identical same-day debits and unannounced bank charges/penalties.
- **Rules 4, 5 & 6 (Outliers & B2B):** Flags statistical spikes (>2.5σ), round-number cash drains, and contract rate breaches.

5. Core Project Architecture & Security Guarantee

Modular Codebase Reference: `app.py` (UI, Universal CSV Mapper & RAM Purge) • `parser_engine.py` (3-Tier PDF & OCR Extraction) • `categorizer_and_auditor.py` (2-Tier Categorization & 6-Rule Auditor) • `excel_exporter.py` (Multi-Sheet Excel & Tally XML Compiler).

Zero-Telemetry Guarantee: Outbound network calls are monkey-patched and blocked at startup. Sensitive client statements never leave the local machine and are wiped from RAM with one click.